Last updated in Sept 2026

Elhanan Buchsbaum

Software & Data Systems · Technical Project Delivery · AI-Assisted Development

Bonn, Germany | ebuchsbaum@protonmail.com | +49 1514 6269424 | github.com/elhananby
linkedin.com/in/elhanan-buchsbaum-2461a7429 | orcid.org/0000-0003-3698-6345 | EU work authorization

Summary

Computational scientist (PhD) who builds software and hardware systems end to end — defining requirements and acceptance criteria up front, implementing and validating the solution, then documenting and training others to run it independently. Nine years in Python and Rust across data pipelines, HPC infrastructure, computer vision, and AI-assisted development (Claude Code, MCP-based tool integrations). Modernised legacy tooling that was then adopted team-wide. Two first-author Current Biology papers. Native English speaker; German B1–B2.

Experience

Doctoral Researcher, MPI for Neurobiology of Behavior (caesar) / University of Bonn – Bonn, Germany — 2019 – 2025

- Designed and built a closed-loop measurement platform for freely flying Drosophila end to end; defined datasets, test setups, and acceptance criteria up front, then implemented multi-camera imaging, real-time 3D tracking, optogenetic control, and microcontroller triggering at 10 ms closed-loop latency.
- Evaluated Python against the platform's timing requirements, found it insufficient, and rewrote the acquisition/ triggering layer in Rust; validated the multi-camera 3D calibration stack quantitatively.
- Built reproducible Python pipelines processing hundreds of recordings in HDF5/Parquet, run in parallel locally and as Slurm batch jobs; open-sourced the platform as OptoFly (github.com/mpinb/optofly, GPL-3.0).
- Documented the system and shipped self-service tools so lab members without programming backgrounds could run analyses independently; trained and supported them; supervised one MSc student.
- **Result:** first-author paper in Current Biology (2025) on a descending neuron driving visually evoked flight saccades.

MSc Researcher, Technion – Israel Institute of Technology, Faculty of Medicine – Haifa, Israel — 2016 – 2018

- Built MATLAB signal-processing pipelines for in vivo electrophysiology and statistics; modernised the lab's legacy analysis codebase and trained incoming researchers — adopted lab-wide. Also taught Python/ML fundamentals to non-programmer medical students (TA). Co-first-author paper in Current Biology (2021) on head-direction coding in the avian hippocampal formation.

Skills

AI & automation: Claude Code, MCP-based tool/skill integration, agentic coding harnesses (opencode, pi), OpenAI-compatible APIs, OpenRouter

Software, data & infrastructure: Python (primary), Rust, MATLAB, R, Bash, SQL, Git/GitHub, NumPy/pandas/scikit-learn, HDF5/Parquet, Linux, Docker, HPC (Slurm)

Computer vision & hardware: OpenCV, FFmpeg, camera calibration, object detection (YOLO), closed-loop real-time instrumentation

Working methods: Translating requirements into acceptance criteria; iterative, incrementally validated development; technical documentation for non-technical audiences; training and onboarding; MS Office (Word, PowerPoint)

Languages: English (native), Hebrew (native), German (B1–B2, actively studying)

Education

University of Bonn / MPI for Neurobiology of Behavior (caesar), PhD in Neuroscience (magna cum laude) – Bonn, Germany — 2019 – 2025

Technion – Israel Institute of Technology, MSc in Neuroscience (summa cum laude) – Haifa, Israel — 2016 – 2018

University of Haifa, BSc in Biology and Psychology – Haifa, Israel — 2013 – 2016